CHAPTER 2

THEORIES OF LANGUAGE CONTACT

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2.1 Introduction

The field of contact linguistics has progressed very rapidly, particularly since Weinreich's (1953) pioneering attempt to formulate a unified interdisciplinary approach to the study of language contact. The roots of the field go back to nineteenth-century studies of various kinds of contact situations and their outcomes, including the foundational work of Hesseling (1899, 1905), Schuchardt (1882, 1883), and others on creoles, and the early attention to linguistic areas such as the Balkan Sprachbund (Kopitar 1829; Schuchardt 1884). Further foundations of the field were laid in early twentieth-century studies of processes of pidgin formation (Broch 1927); contact phenomena such as code-switching (Braun 1937), language maintenance and shift (Kloss 1927); immigrant languages (Herzog 1941, Reed 1948, etc.), and so on.

All of these various lines of approach, some primarily linguistic, others primarily sociological or anthropological, contributed to the emergence of the new field now known as contact linguistics. The field has always been conceived of as a multidisciplinary area of study, built around linguistic, sociolinguistic, and psycholinguistic approaches. Weinreich (1953) was the first to propose a systematized and integrated framework within which language contact could be investigated. In particular, he emphasizes that the components of an explanatory framework must include "purely structural considerations . . . psychological reasons . . . and socio-cultural factors" (1953: 44). These three components—the linguistic, the psycholinguistic, and the sociolinguistic—remain central to the study of language contact. The challenge facing the field is to integrate them into a unified theoretical framework.

A unified theory of language contact must address, first, the nature of the processes underlying contact-induced change, that is, both the actuation and the implementation of change (Weinreich et al. 1980), which relate, respectively, to the roles played by the individual and the community in the origin and spread of change. It is generally agreed

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that the locus of an innovation is the individual; hence the mechanisms or processes of change have to be explained in terms of how linguistic systems or inputs interact in the individual mind (i.e., in psycholinguistic terms). At the same time, if the locus of the propagation of change is the set of networks that link individual to individual, and each to the broader social structure, then this aspect of change must be explained in sociolinguistic terms (Bachus 2009). Meanwhile, the task of linguistic approaches is to describe the kinds of changes that occur, explain their structural characteristics, and identify the principles and processes they involve. Achieving a unified account of these various aspects of contact-induced change is of course an extremely difficult task, so my main goal here is to outline the dimensions of the challenge, by evaluating the contributions made by existing frameworks, and suggesting ways in which they have contributed toward the goal of a unified theory. In section 2.2, I discuss the contributions of sociolinguistic approaches to our understanding of the ways in which social contexts and social factors influence the outcomes of language contact. In section 2.3, I evaluate various frameworks that have been proposed for describing the linguistic outcomes of language contact, the mechanisms involved, and the classification of contact phenomena. I argue that van Coetsem's (1988, 2000) model of language contact offers a more consistent, accurate, and principled explanation of the processes of change associated with different types of contact. In section 2.4, I show how a wide variety of contact phenomena can be accounted for in terms of just two universal mechanisms of change—borrowing and imposition. Section 2.5 discusses ways in which psycholinguistic models of language production can contribute to our understanding of these mechanisms of contact-induced change.

2.2 Social Factors and Language Contact

The centrality of social factors to the understanding of contact-induced change was well recognized by Weinreich.

> To predict typical forms of interference from the sociolinguistic description of a bilingual community and a structural description of its languages is the ultimate goal of interference studies. (Weinreich 1953: 86)

Similarly, Thomason and Kaufman (1988) assign primary importance to the role played by social factors in shaping the consequences of contact, declaring that "[i]t is the sociolinguistic history of the speakers, and not the structure of their language, that is the primary determinant of the linguistic outcome of language contact" (1988: 35). What this suggests, first, is that we need to distinguish among the various social contexts of language contact if we are to understand the nature and direction of cross-linguistic influence. Second, it is necessary to examine, where possible, the actual

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speech behavior of individuals in each contact situation in order to uncover the factors that motivate them to change their language in one way or another. Weinreich's insight also led him to distinguish the non-structural factors that operate on both the macro (societal) level, and those that operate on the micro (individual level).

The first challenge for a sociolinguistic approach is to determine how the overall organization of the speech community affects the outcomes of language contact. The issue here is whether differences in the configuration of social groups and their languages correlate with differences in the types of contact-induced change. Scholars have proposed different taxonomies of speech communities in this regard. For instance, Loveday (1996: 16) has suggested that communities might be categorized according to the degree of bi- or multilingualism within them. He suggests that there are six "archetypal contact settings," each characterized by different arrays of contact phenomena, which form a continuum from relatively monolingual to highly multilingual. At one end of the spectrum we find relatively homogeneous communities of monolinguals, most of whom have little or no direct contact with speakers of other languages. Such "distant" contact typically results in lexical borrowing alone. In the middle, there are situations involving varying degrees of bi- or multilingualism within the community. These include communities with minority language groups that are either relatively isolated (e.g., Gaelic in Scotland), or in which different ethnic groups vie for equal status in the same territory, each preserving its own language, but also learning the other (e.g., French and English in Montréal). Then there are communities characterized by "diglossia," a situation in which two languages, one (H)igh and the other (L)ow, fulfill complementary functions in the community (e.g., Standard German and Schwyzertüütsch in Switzerland). Many situations in the middle of the spectrum are characterized by ongoing language shift, particularly among minority groups subject to strong cultural pressure from a dominant group. At the other extreme of the continuum, we find highly heterogeneous communities characterized by high degrees of individual multilingualism, such as the village of Kupwar in India, described by Gumperz and Wilson (1971).

Other contact situations involve different speech communities that engage in constant interaction, and the fluidity of their social boundaries is matched by the fluidity of their linguistic practices. The Aboriginal groups of Arnem Land, Australia (Heath 1978) and the villages of Northwest New Britain in Papua/New Guinea (Thurston 1987, 1994) are examples of this type. All of these settings offer a rich range of possibilities for contact-induced changes of one type or another. There may be borrowing across languages, code-switching behaviors, substratum influence on varieties acquired as second languages, various types of convergence, and the creation of entirely new contact languages. The particular outcomes have to do with a range of social factors, some favoring the preservation of language boundaries, others favoring different degrees of language mixture, switching, and convergence, yet others promoting language shift.

Finally, there are the social contexts that lead to the formation of new contact languages, including pidgins, creoles, "indigenized varieties," and bilingual mixed languages. Each of these outcomes involves a different constellation of social groups, interactional settings, and social relationships. The contexts in which they arise have

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been well documented, so there is no need to repeat them here. An interesting approach, which attempts a unified account of how macro-level social factors shape the creation of creoles and indigenized varieties, is Mufwene's (2001, 2008) Ecology of Language (EL) framework. Mufwene argues that different colonization types, and the kinds of economic systems they engendered, led to differences in population structures, which directly affected the evolution of English and other European languages in their colonial settings. Thus, in general, colonies established mostly for trade purposes in places like Africa and the Pacific produced pidgins, while "exploitation" colonies in the same regions produced "Indigenized" varieties such as Nigerian, Indian, and Singapore English. By contrast, "settlement" colonies such as those in the United States, Australia, New Zealand, and so on, produced outcomes that were closely related to their European sources, while "plantation" colonies produced creoles. Mufwene and Vigouroux (2013: 111ff) identify a number of macro-level social factors that influenced language evolution in these colonial settings, including the following:

- Differences in economic systems (which influence population structure);
- Population structures (which determine who interacts with whom);
- Demographic strengths of populations.

The proposed taxonomy has much in common with those proposed by scholars like Gupta (1997) and Schneider (2010). Mintz (1971: 48) proposes a similar model of the three broad social conditions that shaped the emergence of creole languages in the New World:

- The demographic makeup of each colony, including population ratios between the groups, and their places of origin;
- The types and patterns of contact among the groups, which were generally determined by the codes of social interaction governing their relative statuses and relationships;
- The nature of the community settings in which the groups interacted.

These macro-level factors figure in most frameworks for investigating how contact languages emerged.

Though these taxonomies are quite useful, it is clear that there is no consistent correspondence between the type of community and the pattern of contact-induced change within it. Bilingual communities, for instance, may be characterized by stable maintenance in some cases, language shift in others, or by both. Long-term stability can translate into rapid shift, given the right circumstances. Each contact situation has its own set of social contexts and forces at work, which determine whether individuals or groups are more or less proficient in one or the other language, whether they maintain their ancestral languages or shift toward a second language, or whether they maintain a kind of balanced bilingualism. Each of these sociolinguistic configurations can have profoundly different effects on how the languages in contact influence each other, since they determine not just the social dominance relationships between the languages, but also their linguistic dominance relationships within individual members of the community.

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The second major task of sociolinguistic approaches is to explain the processes by which innovations spread, or competition between alternative features is resolved at the level of the community. This calls for closer examination of the social factors that operate at the micro-level of individual language use and choices, which in turn involve the patterns of interaction that individuals engage in. Mufwene's EL framework attaches central importance to such micro-level factors in regulating the spread of changes across idiolects. In this approach, the propagation of changes at group or community level involves a process of selection from among competing variants—a process that is subject to various external constraints, pertaining to both social and linguistic influences. This model of competition and selection at both individual and group level lends itself quite readily to an explanation of change in sociolinguistic terms. The approach is by no means new, and has been explored in numerous ways in the literature. In particular, sociolinguists have investigated the propagation of change within and across social networks (Milroy and Milroy 1985; Eckert 2000; Labov 2001); the role of "leaders" of change (Eckert 2000; Labov 2001); and the role of accommodation or divergence in promoting or inhibiting change (Giles et al. 1991). Weinreich once more anticipated this approach in stressing that the individual speaker's "specialization in the use of each language by topics and interlocutors" was an important factor in contact-induced change. In addition, he identified the following factors (1953: 3–4):

- The speaker's facility of verbal expression in general and his or her ability to keep two languages apart;
- Relative proficiency in each language;
- Manner of learning each language;
- Attitudes toward each language, and whether idiosyncratic or stereotyped.

As we will see in section 2.5, the first three of these factors are quite relevant to understanding the psycholinguistic mechanisms underlying contact-induced change.

We are still a long way from fully understanding the ways in which micro-level social factors interact with others in regulating the outcomes of language contact. This represents a promising area for future research (see the papers in Leglise and Chamoreau 2013 for a step in this direction).

2.3 Classifications of Contact Phenomena, and Mechanisms of Change

One of the major obstacles to a unified theory of language contact has been the failure of linguists to achieve consensus on two broad issues: the definition and classification of contact phenomena, and the processes and principles underlying different types of contact-induced change. With regard to the former, most frameworks follow

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Weinreich's distinction between contact-induced changes due to borrowing, and those due to what Weinreich called "interference." For instance, Thomason and Kaufman (1988: 37) classified contact-induced changes into two broad types—those due to borrowing, and those due to interference through shift (1988: 37). Other scholars have adopted different terminology. Thus Johanson (2000, 2002) distinguishes between "adoption" (a term that he prefers to "borrowing") and "imposition," which corresponds closely to interference through shift (see further discussion later in the chapter). Unfortunately, there are various problems with these classifications. In the first place, there is by no means any agreement on what terms like "borrowing" and "interference" refer to. For instance, Aikhenvald (2002), following Trask (2000: 44), defines borrowing as "the transfer of features of any kind from one language to another as the result of contact." For Aikhenvald, apparently, borrowing covers a broad range of phenomena, including cases of "direct diffusion," that is, the transfer of overt forms, as well as "indirect diffusion," the transfer of categories and patterns (Aikhenvald 2002: 4). Similarly, some scholars use the term "interference" to refer to any type of cross-linguistic influence, including borrowing, while others use "transfer" in the same broad sense. Andersen (1983: 7) discusses the "long and confusing history" of these terms.

To complicate matters further, some scholars employ classifications that are based primarily on the nature of the linguistic features that are transferred. For instance, Heath (1984: 367) distinguishes between "direct transfer of forms from the other language," and "structural convergence (using native morphological material)." Croft (2003: 51) distinguishes between "borrowing" and "convergence," describing the former as the transfer of "substance linguemes" (i.e., actual forms with their meanings), and the latter as the transfer of "schematic linguemes" (i.e., grammatical elements which are made up of form only or meaning only, but not the combination of the two). Matras and Sakel (2007: 829) distinguish between "replication of linguistic matter" or MAT (i.e., morphemes and phonological shapes from a source language), and "replication of patterns" or PAT (i.e., patterns of distribution of grammatical and semantic meaning, and formal-syntactic arrangement). Heine and Kuteva distinguish between borrowing, which they define as "contact-induced transfer involving phonetic substance of some kind or other" (2005: 6), and the transfer of meanings (including grammatical meanings or functions) and of syntactic relations (i.e., the order of meaningful elements) (2005: 2). Finally, Johanson distinguishes between "global copying" (the total transfer of a linguistic element, including its form and functions) and "selective copying" (the copying of selected structural properties without the material forms of the elements in question) (2002: 291).

All of the preceding classifications are problematic because they refer primarily to the results of contact-induced change, not to their underlying mechanisms or psycholinguistic processes. Van Coetsem identifies the problem precisely when he notes that the terms "borrowing" and "transfer," as they are traditionally used, "do not indicate the direction of the influence, and thus fail to bring out the agent of the action" (van Coetsem 1988: 2). Van Coetsem proposes a classification that is based precisely on this notion of agentivity, which I discuss more fully in the following subsection, as a prelude to evaluating other current frameworks.

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2.3.1 Van Coetsem's Framework

Like other researchers, van Coetsem makes a broad distinction between two broad mechanisms of contact-induced change, which he calls "borrowing" and "imposition," but his main contribution was to refine and clarify the traditional distinction between the two mechanisms by making more explicit the types of agency that they involve. In both types of transfer, there is a source language and a recipient language. The direction of transfer is always from the source language to the recipient language, but the agent of the transfer can be either the recipient language speaker or the source language speaker. In the former case we have borrowing (recipient language agentivity); in the latter, imposition (source language agentivity). The distinction between these two types of transfer is based, crucially, on the psycholinguistic notion of language dominance. This refers roughly to the degrees of proficiency that the speaker has in each language, though it allows for the fact that a speaker may have different degrees of proficiency in different areas of a language. Generally, however, a speaker is linguistically dominant in the language in which he is more proficient or fluent—which is usually, but not necessarily, his first or native language (van Coetsem 1988: 13). van Coetsem (1988: 3) defines borrowing as follows:

> If the recipient language speaker is the agent, as in the case of an English speaker using French words while speaking English, the transfer of material (and this naturally includes structure) from the source language to the recipient language is *borrowing (recipient language agentivity)*. (italics in original)

Imposition, on the other hand, is a process by which the speaker transfers features of her linguistically dominant language (as SL) into her version of the recipient language (RL). In this case, "the source language speaker is the agent, as in the case of a French speaker using his French articulatory habits while speaking English" (van Coetsem 1988: 3). There are several aspects of van Coetsem's framework that are crucial to our understanding of the processes underlying contact-induced change. One is the assertion that contact-induced changes originate as innovations in individual speakers' language use. This reflects a common understanding among contact linguists, which is reflected in Weinreich's (1953: 6) observation that "the individual is the ultimate locus of contact." A second major premise of van Coetsem's model is that such changes are introduced in the course of language production, that is, in "the speech behavior of bilingual individuals" (Weinreich 1953: 4). But the most important of van Coetsem's claims is that it is the nature of the linguistic dominance relations between the languages in contact that decides the agency and type of contact-induced changes. This is a radical departure from other approaches that assigned primary importance to the social dominance relationship between the languages in contact. It must therefore be emphasized that linguistic dominance is an individual psycholinguistic phenomenon, while social dominance is a sociopolitical concept, based on the power or prestige standing of one of the languages. Only the former is relevant to the agency of contact-induced change and its underlying processes and mechanisms. It must be noted also that the socially

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dominant language may or may not be the linguistically dominant language of the speaker as agent of change. In addition, both linguistic and social dominance relationships may change over time, in individual speakers and in the community at large. Such reversals of dominance relationships also play a crucial role in determining the outcomes of language contact, but they have largely been ignored in the literature, as we will see. It is precisely these insights of van Coetsem's framework that are lacking in other approaches to contact-induced change, which we now turn to.

2.3.2 Competing Frameworks for the Study of Contact-Induced Change

One of the most important contributions to our understanding of contact-induced language change was Thomason and Kaufman's (1988) study, which provided a highly innovative historical linguistic perspective on language contact and its outcomes, and established the foundation for future work in the field. They emphasized the interaction of linguistic and social factors, and the key role played by the sociolinguistic history of the community, in determining the nature and degree of contact-induced change. Their classification of contact-induced changes is based on a distinction between "borrowing" and "interference via shift." However, the framework is highly problematic in some respects, primarily because of vagueness or inaccuracy in their definitions of the two key mechanisms of change.

Thomason and Kaufman (1988: 37) define borrowing as "the incorporation of foreign features into a group's native language by speakers of that language: the native language is maintained but is changed by the addition of the incorporated features." The other mechanism—"interference via shift"—refers to the transfer of features of learners' first languages into a second language that they are in process of acquiring. While this distinction is helpful in identifying a wide variety of contact-induced changes, it suffers from serious flaws when applied to certain situations of language contact. The chief weakness in Thomason and Kaufman's approach is their assumption that borrowing is inextricably linked to language maintenance, and that "interference" is exclusively a product of shift that produces "imperfect" second-language acquisition. In fact, there is no one-to-one relationship between any mechanism of change and the type of language situation that is involved. This is because both situations of maintenance and situations of language shift can involve differences in dominance relationships between the languages in contact. Hence there is potential for both borrowing and imposition to come into play in both situations. Moreover, linguistic dominance relations vary from individual to individual, and from community to community. This makes it impossible for us to treat all cases of language maintenance or shift as if they were the same, or could produce predictable results.

Among the situations that pose particularly difficult problems for Thomason and Kaufman are those in which a community maintains its ancestral language while becoming bilingual, and increasingly more proficient, in a socially dominant language. Such a

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situation existed in the immigrant Dutch community of Iowa, as described in Smits (1998), where English had become the linguistically dominant language for most speakers, though Dutch was still maintained. As a consequence, many speakers are no longer very proficient or fluent in their ancestral language. The linguistic consequences include the following:

- Lexical influence from English is highly marginal;
- The Dutch inflectional system has been seriously affected both by internal processes of reduction and external influence from English;
- There is a great deal of phonological and syntactic interference from English.

Smits concludes from this that the changes in Iowa Dutch are clearly the result of imposition, that is, transfer from the linguistically dominant language, English, into the less dominant one, Dutch, under the agency of English-dominant bilinguals. Indeed, these changes are precisely what van Coetsem's model would predict. Yet, under Thomason and Kaufman's approach, such changes would have to be ascribed to borrowing. Not surprisingly, then, Smits criticizes Thomason and Kaufman's approach for including under the umbrella of borrowing both changes that are due to RL agentivity, and others that are due to SL agentivity or imposition. She notes that, in their approach,

> [. . .] borrowing not only entails interference from a linguistically non-dominant language into a maintained *linguistically dominant* RL . . . but also entails interference from a linguistically dominant SL into a *linguistically non-dominant* RL.
> (1998: 387; italics in original)

As a consequence, Thomason and Kaufman's notion of borrowing "refers to two fundamentally different contact situations, yielding fundamentally different linguistic effects" (Smits 1998: 387).

For these reasons, Thomason and Kaufman's interpretation of "borrowing" leads to mischaracterization of a broad range of contact-induced changes that occur in situations of language maintenance, where a community maintains its ancestral language while shifting to a second language. When such speakers attain greater proficiency or even dominance in their second language, they tend to transfer features from it to their original language. Such changes fall under the ambit of imposition via SL agentivity. These are the kinds of changes that have been mistakenly characterized as cases of structural borrowing in the work of Thomason and Kaufman as well as others. A well-known case in point is that of Asia Minor Greek, which changed drastically at all levels of structure under the influence of Turkish. Thomason and Kaufman (1988: 218) treat this as a case of structural borrowing, arguing that "if Turks did not shift to Greek, all of the interference must be due to borrowing." Winford (2005: 408) notes that "this overlooks the strong possibility that bilinguals, especially those that were Turkish-dominant, played a key role in introducing these changes." This is in keeping with van Coetsem's observation that "the linguistic dominance relation between the RL and the SL [. . .] determines whether RL or SL agentivity will result from the contact" (1988: 83).

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Thomason (2003: 692) revises her distinction between borrowing and shift-induced interference, arguing that "the crucial sociolinguistic factor is not whether or not shift takes place, but whether or not there is imperfect learning by a group of people." This leads her to redefine borrowing as transfer into a language spoken fluently, and "shift-induced interference" as transfer into a language that is learned imperfectly. But, as Smits (1998: 386) notes, the reformulation still mischaracterizes as "borrowing" changes that are introduced into a maintained language by speakers who have become dominant in a second language—changes which, as we have seen, are due to imposition.

Another framework that has gained currency in recent times is that of Johanson (2002), who makes a clear distinction between the results of contact-induced change and the mechanisms underlying them. He uses the term "code copying" to refer to the resulting contact phenomena, which he classified into two broad categories—global and selective code copying (2002: 291). The former refers to the copying of both the form and functions of some linguistic element, while the latter type refers to copying of structural features, and "is traditionally known as 'loan phonology,' 'loan semantics,' 'loan syntax,' etc." (2002: 292). According to Johanson, code copying is the result of two mechanisms, which he labels "adoption" and "imposition." Adoption refers essentially to the mechanism that others refer to as borrowing—a term that Johanson rejects as being "based on a deceptive metaphor" (2002: 288). By contrast, Johanson's concept of "imposition" is fundamentally different from van Coetsem's. This is because Johnason defines both borrowing and imposition in terms of the social dominance relationship between the languages in contact, not in terms of linguistic dominance, as in van Coetsem's framework:

> In the case of adoption, speakers of a sociolinguistically-dominated code A insert copies from a sociolinguistically-dominant code B. In the case of imposition, speakers of the sociolinguistically-dominated code A insert copies from it into their own variety of the sociolinguistically-dominant code B. (2002: 290)

This strict equation of transfer type with a particular social dominance configuration of course runs completely counter to the psycholinguistically based association of a transfer type with a particular linguistic dominance configuration, as in van Coetsem's model. For Johanson, any form of transfer from a socially dominant to a subordinate language must be borrowing, while any form of transfer in the opposite direction must be imposition. This means that, like Thomason and Kaufman, his classification of contact-induced changes is flawed, and leads to inaccurate predictions for many cases of language contact. For instance, he cites Jones's (2002) description of changes in Guernésiais (Guernsey French) under English influence as instances of "adoption" (borrowing, in our sense). Examples provided by Jones include the following:

- The tendency to prepose adjectives;
- Copying of the syntax of passive constructions;
- Use of a single preposition 'with' for both comitative and instrumental meanings;
- Change of meanings and connotations of Guernésiais verbs due to formal similarities to English verbs.

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Again, such cases of transfer can be better explained in terms of the growing dominance of English among speakers of Guernésiais, leading to imposition (in our sense) of the same sort that Smits describes for Iowa Dutch.

Johanson himself has nothing to say about linguistic dominance relationships and the role they play in contact-induced change. In fact, he specifically eschews any attempt to explain contact phenomena in psycholinguistic terms:

> [...] changes due to code-copying will be discussed exclusively in terms of observable linguistic structures. It will not be claimed that copies are psycholinguistically produced or processed in the steps discussed. (2002: 287).

Such an approach is unfortunate, since it incorrectly assigns many instances of contact-induced change to the wrong category of "adoption" (i.e., borrowing), rather than recognizing them as cases of imposition, in van Coetsem's sense of the term. The implications are far-reaching, since imposition can occur in any situation of bi- or multilingualism where the dominance relationship between the languages changes. As we will see in section 2.4.2, contemporary bilingual situations provide rich and abundant illustrations of these patterns of change.

None of the preceding criticism is meant to deny the importance of social dominance relationships between languages as an important factor in contact-induced change. Rather, it is meant to remind us that such relationships are often trumped by the nature of the linguistic dominance relationship between the languages in contact among the speakers who are the agents of change. Moreover, since it is based on how linguistic dominance shapes the agency of speakers, van Coetsem's framework is compatible with psycholinguistic approaches to language contact (see section 2.5). His emphasis on the cognitive processes involved in the creation of contact languages is an important complement to other approaches that are more concerned with sociohistorical and sociolinguistic aspects of contact. In addition, van Coetsem's model has the distinct advantage of uniting a wide variety of apparently disparate contact phenomena under the umbrella of each of the major mechanisms of change. This is discussed in the following section.

2.4. On the Unity of Contact Phenomena

Each of the major mechanisms of contact-induced change, borrowing and imposition, is associated with a wide variety of contact phenomena that have been treated independently of one another in the literature. We will first discuss those that can be unified under the umbrella of borrowing.

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2.4.1 Outcomes of Borrowing

To recap, *borrowing* is defined as the transfer of linguistic materials from a source language into a recipient language via the agency of speakers for whom the latter is the linguistically dominant language, in other words, via RL agentivity. Essentially, it refers to a psycholinguistic mechanism by which speakers introduce materials from an external language into a language in which they are (more) proficient. In doing so, they tend to preserve the more stable (i.e., structural) domains of the recipient language. This is not to say that borrowing never involves structural elements or abstract grammatical features, only that such structural borrowing occurs only under specific circumstances, which will be discussed further later. It follows that most borrowings tend to be lexical, and it is not surprising that the bulk of the literature has been devoted to these. Such borrowing does not lead to simple copying of source language materials, since various creative processes of adaptation and integration come into play as part of the mechanism. Thus the results of lexical borrowing are quite varied, ranging from close imitations of SL items (e.g., *rendezvous* borrowed from French into English), to drastic adaptations such as Japanese *pokemon* (<English *pocket monster*), to imitation of compounding patterns (Spanish *rascacielos* modeled on *skyscraper*), to completely new creations based on native materials (Zapotec *éxxuwí* 'fig' < *exxu* 'avocado' + *wí* 'guava'), and so on. The various processes of imitation and adaptation exemplified here demonstrate that borrowing involves complex patterns of lexical change that create new lexical entries or modify existing ones. In all cases, borrowed items are manipulated so that they conform to the structural and semantic rules of the recipient language. This is the definitive mark of borrowing under recipient language agentivity.

In addition, lexical borrowing often acts as the conduit for the introduction of new structural features, both phonological and morphological, into a recipient language. For instance, distinctions between /s/ and /z/, /f/ and /v/, and /ʃ/ and /ʒ/ emerged in Middle English as a result of the heavy borrowing of French words containing the voiced members of the respective pairs (Thomason and Kaufman 1988: 124, 308). But see Laker (2009) for an alternative explanation. At the morphological level, free functional elements such as conjunctions, prepositions, pronouns, and even complementizers can be borrowed directly. On the other hand, bound derivational morphemes tend to be introduced to a recipient language along with lexical borrowings, and can become productive if the borrowed words are numerous. For example, French borrowings into Middle English, such as *conspir-acie*, *charit-able*, and others, yielded suffixes like *–acy* and *–able*, while *en-rich, dis-connect*, and others yielded new prefixes (Dalton-Puffer 1996). Direct borrowing of overt inflectional morphology appears to be very limited; again, however, lexical borrowing can be the channel for such transfer, as exemplified in the introduction of the plural inflection *–im* into Yiddish via borrowed Hebrew pairs such as *mín/minim* 'sort' (Thomason and Kaufman 1988: 21).

2.4.1.1 *Lexical Borrowing Compared with Other Contact Phenomena*

The view of borrowing as a process that involves recipient language agentivity allows us to link phenomena that have been interpreted in very different ways in the literature.

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Among them are classic code-switching,[1] relexification, and the creation of bilingual mixed languages. In classic insertional code-switching, a speaker retains the morphosyntactic frame of his dominant language (as the RL), and imports single content morphemes or phrases from an external source language, as in the following example of French/Arabic switching (Bentahila and Davies 1983: 319).

(1) C'est une pauvre *bint*.
'She is a poor girl.'

Myers-Scotton (1993, 2002) has shown that, in these cases, one language, the RL in our terms, acts as the matrix language, that is, provides the morphosyntactic frame for the utterance. Moreover, the items incorporated from the embedded or source language consist mainly of open-class content morphemes, which are inflected according to the rules of the recipient language. For this reason, it seems reasonable to treat insertional code switches and lexical borrowings as manifestations of the more general phenomenon of borrowing under recipient language agentivity (Winford 2009). While the results may differ in some ways, the underlying mechanism is the same. Understanding borrowing in terms of processes rather than results also helps us to explain how certain kinds of contact language were created in the past. In some contact situations, the process of incorporating content morphemes into a recipient language can be taken to an extreme, resulting in new languages that derive their morphosyntactic frame from one language, and their lexicon from another. These creations are referred to as intertwined languages, or "Lexicon-Grammar mixed languages" (Bakker 2003). Perhaps the best-known example is Media Lengua, a language spoken in central Ecuador, in which almost the entire grammatical frame is Quechua, but practically all content morphemes or lexemes derive from Spanish. The following example illustrates (Spanish items in italics):

(2) Media L: *Unu* *fabur*-ta *pidi*-nga-bu *bini*-xu-ni (Muysken 1997: 365)
one favor-ACC ask-NOM-BEN come-PROG-1
'I come to ask a favor.'
Quechua: Shuk fabur-da maña-nga-bu shamu-xu-ni
Span: Vengo para pedir un favor.

The process of incorporating and integrating Spanish stems into a Quechua morphosyntactic frame is identical to what we saw earlier in the cases of classic code-switching. Muysken described this process as "relexification . . . the process of vocabulary substitution in which the only information adopted from the target language [source language – DW] in the lexical entry is the phonological representation" (Muysken 1981: 61). It seems clear that the process of lexical incorporation that Muysken describes is in essence the same as that which occurs in lexical borrowing

[1] Myers-Scotton (2002) uses this term to refer to the insertion of embedded language morphemes into the morphosyntactic frame of a matrix language.

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and classic code-switching. In all cases, source language lexical forms are imported and integrated into the unchanged structural frame of a recipient language, via RL agentivity. Further discussion of the psycholinguistic aspects of this process can be found in section 2.5.

2.4.2 Contact Phenomena United under the Umbrella of Imposition

The mechanism of imposition offers a more consistent, accurate, and principled explanation for various types of grammatical replication associated with different scenarios of language contact, including second-language acquisition (SLA), creole formation, language attrition, and others. I use "grammatical replication" (Heine and Kuteva 2005: 6) as a cover term for a wide range of labels that have been used in the literature, including pattern replication, indirect diffusion, functional transfer, selective copying, substratum transfer, convergence, and so on. Labels like these seldom speak to the actual psycholinguistic processes that initiate change as a result of the contact between competing linguistic systems in the minds of individual speakers who are the agents of change. All of the instances of grammatical replication mentioned in the preceding involve transfer from a linguistically dominant first language into a non-dominant second language, so they are clear cases of imposition or SL agentivity. They fall into two broad types. First, there are cases that involve the replication of some L1 grammatical pattern, meaning, or function in a language acquired as an L2; Liu et al. (1992: 451) refer to this as "forward transfer." Second, there are cases that involve transfer of patterns from an L2 in which the speaker has become dominant, into an original L1 that is undergoing attrition; Liu et al. (1992: 451) refer to this as "backward transfer," that is, "the use of L2 processing strategies in L1, a possible symptom of language loss."

2.4.2.1 *Imposition from L1 to L2*

The vast majority of the literature is concerned with transfer from L1 to L2, which covers situations involving tutored and natural SLA, as well as (expanded) pidgin and creole formation. There is a huge literature on these kinds of change, so I will provide only a few examples here, from situations of natural SLA and creole formation. An example of the former type of situation is Andean Spanish. Odlin (1990) reports that, in this contact variety, the SVO order of general Spanish is often replaced by an SOV pattern, under the influence of Quechua, as in the following example (Odlin 1990: 103):

(3) Y mi hermano aquí otro paloma hembra había chapado
'And my brother here another dove female had caught.'

Similarly, Sanchez (2006) discusses the Andean Spanish use of the Spanish verb *querer* 'to want' to convey the sense of an imminent action, on the model of the Quechua affix

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–*naya*-, which expresses both a desiderative sense, and the imminent nature of a verbal action. The following examples from Sanchez (2006: 289), produced by the same bilingual child, illustrate:

(4) And. Span. *Y el perrole (e)sta queriendo morder a ese sapo*
And the dog is wanting to.bite P that toad
'And the dog wants to/is about to bite the toad'

(5) Quechua *Achku miku-naya-yka-n*
Dog eat-DES-PROG-3
'The dog wants to/ is about to bite.'

Similar kinds of impostion are found in creole formation. For example, Caribbean creole languages have a wide range of serial verb constructions that replicate similar structures in the West African substrate languages. For instance, Migge (1998) demonstrates that Ndjuka, a Surinamese creole, employs a range of SVCs with "give" as the V2, where 'give' introduces a range of thematic roles (recipient, benefactive, substitutive, etc.) that are also found in Gbe and other languages that were the principal substrates for the Surinamese creoles.

(6) Ndjuka *Mi seli a osu gi en* (Migge 1998:236)
1sg sell DET house SV 3sg
'I sold the house to him/her'

(7) Ewegbe *ye dʒra maʃina-a ne amba*
3pl sell machine-the SV Amba
'They sold the computer to Amba'

2.4.2.2 *Imposition from L2 to L1*

Unlike cases of transfer from L1 to L2, transfer in the other direction has been largely neglected in the contact linguistics literature, even though there is evidence that it is quite common. Such "backward" transfer is particularly common among bilinguals for whom the L2 has become the primary and dominant language, and whose L1 is undergoing attrition. Clyne (2003) discusses various examples of how Dutch and German immigrants in Australia, who have become dominant in English, introduce English syntactic patterns into their declining ancestral languages.

(8) a. Aus. German *Wir haben gegangen zu Schule in Tarrington*
We have gone to school in Tarrington

b. St. German *Wir sind in Tarrington zur Schule gegangen*
we Aux.be in Tarrington to school gone
'We went to school in Tarrington.' (Clyne 2003: 80)

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(9) a. Aus. Dutch *je heb te look voor een ander job*
b. St. Dutch *je moet een andere baan gaan zoeken*
you must an other job go seek
'You have to look for another job.' (Clyne 2003: 178)

Other examples of this kind of "backward" transfer include the cases of Guersney French and Iowa Dutch, which were discussed earlier. Silva-Corvalán (1998) also describes changes in the Spanish of third-generation Hispanic immigrants in Los Angeles, which are due to influence from their more dominant language, English.

This brief summary shows that grammatical replication via imposition is an extremely common phenomenon in various contact situations in which one language dominates the other in bilinguals' competence and production. It manifests itself in cases of language shift, where L1 features are imposed on a developing L2, as well as in cases of L1 attrition, where features from a now dominant L2 are imposed on the declining L1. In the following section, I discuss how psycholinguistic models of language production can shed light on the nature of borrowing and imposition as the two major mechanisms of contact-induced change.

2.5 Pyscholinguistic Insights into Contact-Induced Change

Psycholinguistic approaches to language contact view contact phenomena as arising from the different contributions made by the languages in contact during the course of language production. Most approaches are based on Levelt's (1989) model of monolingual language production, which has been modified by de Bot (2000) and others to account for bilingual production. The full details of the model need not concern us here (see Bock and Levelt 1994 for further details). It is generally agreed that three types of mental processes are involved: conceptualization processes that specify which concepts are to be expressed verbally; formulation processes that select appropriate lexical items and construct the syntactic and phonological structure of the utterance; and articulation processes that realize the latter as overt speech (Roelofs 1993: 108). The general outline of the model is shown in Figure 2.1 (Bierwisch and Schreuder 1993: 25).

Our main concern here is with the level of the Formulator, which is concerned with grammatical processing or encoding. This in turn involves lexical selection, which "involves identifying the lexical concepts and LEMMAS suitable for conveying the message" (Bock and Levelt 1994: 947), the building of constituent structures via assignment of functions to selected lexical items, and fixing the order of these items, among other procedures.

De Bot (2001) has suggested various ways in which Levelt's model of monolingual production can be adapted to account for bilingual speech, particularly code-switching.

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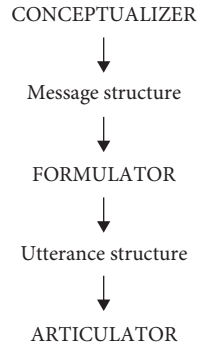

FIGURE 2.1 A simplified model of language production.

Source: Levelt (1989).

As he points out, the model must not only account for cross-linguistic influences of various types, but also "deal with the fact that the speaker does not master both language systems to the same extent" (2001: 425). Hence, such a model must be able to account for the following aspects of bilingual language production (De Bot 2001: 425):

- The two language systems involved can be used independently of each other, or they may be mixed in various ways (as in code-switching, for example);
- The two systems may influence each other;
- The bilingual speaker may have different degrees of proficiency in each system;
- Interactions can take place between languages that are typologically similar or dissimilar.

De Bot accepts Green's (1986) suggestion that there may be three levels of activation involved in plurilingual language production. First, there is a selected language that controls the speech output; second, there is an active language that plays some role in ongoing processing and works parallel to the selected language; and third, there may be a dormant language that plays no role in processing. According to this hypothesis, one language is always selected, but more than one language may be active or latent. Moreover, the active language may do everything the selected language does, such as retrieve lexical items, form sentences, generate surface structures, and even make a phonetic plan (Green 1986). It follows that differences in the roles played by each language in the production process can lead to different kinds of contact phenomena or language mixture. Crucially for our purposes, the nature of these roles will depend on which language is dominant in the speaker's competence, and in the production process.

Let us first consider how this would apply to cases of borrowing and code-switching, that is, recipient language (RL) agentivity. Pyscholinguistic approaches view dominance relations between languages in code-switching in terms of which language is more activated in the production of code-mixed utterances. Myers-Scotton's (2002) notion of "matrix language" (ML) is in fact specifically defined as the language whose grammatical encoding procedures are more activated in the production of mixed

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speech. The ML is equivalent to the RL in van Coetsem's framework. In both lexical borrowing and classic insertional code-switching, speakers employ the morphosyntactic procedures of the RL as the dominant language, and selectively introduce source language (SL) lexical items, or more accurately lexemes, which are associated with lemmas belonging to RL lexical entries. This is basically the same procedure that creators of intertwined languages employed in fashioning languages such as Media Lengua and Angloromani, which Muysken's relexification model attempts to capture. What it suggests is that bilinguals can in fact attach the source language lexeme to the RL lemma as an alternative phonological shape that replaces the equivalent RL lexeme (de Bot 2001: 441). In terms of van Coetsem's framework, then, direct lexical borrowing, insertional codeswitching, and the across-the-board insertion of SL lexemes found in intertwined languages are all the result of the same linguistic mechanism that we have defined as borrowing via RL agentivity.

Insights from psycholinguistic models can also shed light on how the mechanism of imposition operates to produce various kinds of grammatical replication resulting from both "forward" and "backward" transfer. From the psycholinguistic perspective, imposition can be described as the use of the formulation procedures of the SL in producing utterances in the RL. We can illustrate this with two examples of grammatical replication. Recall that a key part of grammatical encoding involves lexical selection, that is, identifying the lexical concepts and lemmas that are suitable for conveying the message. As we saw earlier, one of the assumptions of language production models is that lemmas activate or trigger syntactic procedures that correspond to their syntactic specifications. Thus, a verb will trigger construction of a VP, a noun the construction of an NP, a preposition the construction of a PP, and so on. The argument structure and syntactic subcategorization properties of verbs, in particular, are crucial to appropriate assignment of functions to the constituents of a sentential structure. There is evidence from the SLA literature that many cases of transfer from L1 to L2 involve the assignment of an L1 lemma to an L2 verb form or lexeme, which results in production of L2 utterances that have the grammatical pattern of the L1. For instance, Nemser (1991) provides the following examples of L2 English, produced by L1 German learners in Austria.

(10) L2 English *Explain me something*
German *Erklär mir was*

(11) L2 English *You just finished to eat*
German *Du hast gerade aufgehört zu essen*

In these cases, the L2 English verbs assume the subcategorization properties of their semantically equivalent German counterparts, resulting in utterances that employ English lexical items, organized according to German constituent structure.

Similar processes can be observed in cases of "backward transfer." For example, Silva-Corvalán (1998: 233) has documented examples of Spanish utterances like the

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following, produced by Spanish-English bilinguals in Los Angeles, for whom English has become the dominant language.

(12) a. LA Spanish. *Yo* *gusto* *eso*
I like-1s that

b. Gen. Span. *A* *mi* *me* *gusta* *eso*
To me pro please-3s that
"I like that."

In this case, the argument structure of English *like* is substituted for that of Spanish *gustar*. In General Spanish, the theme of *gustar* is mapped onto the subject function, while the Experiencer role is mapped onto an indirect object. However, the speaker producing the English-inflenced variety of LA Spanish has activated the lemma associated with English *like*, and imposed it on the Spanish lexeme *gustar,* producing an English-like structure in which the Experiencer role is mapped onto the subject function and the Theme role onto the object function. This procedure is motivated by the speaker's perception of a semantic equivalence between *like* and *gustar*—a classic trigger for imposition.

2.6 Other Approaches to Grammatical Replication

Treating cases of grammatical replication in terms of imposition seems intuitively more economical than other approaches that appeal to different mechanisms of change, such as "pivot matching" (Matras and Sakel 2007) or "contact-induced grammaticalization" (Heine and Kuteva 2005). Matras and Sakel suggest that pivot matching "involves identifying a structure that plays a pivotal role in the model construction and matching it with a structure in the replica language, to which a similar, pivotal role is assigned in a new replica construction" (2007: 830). However, this implies that speakers can match entire constructions and then replace one with the other, and it also assumes far greater metalinguistic awareness on the part of speakers who have low proficiency in one of the languages than they actually have. Moreover, Matras and Sakel claim that "pattern transfer" is often excluded when there is no available structure in the replica language that can assume the role of the "pivotal feature" of the model construction (2007: 847). However, this makes little sense for most cases of grammatical replication, particularly those involving learners who attempt to use an L2 with which they are not familiar. In such cases, learners clearly have no knowledge of the grammatical encoding procedures of the L2, far less any knowledge of L2 "structures" that might be equivalent to those of their L1. Yet they manage to replicate quite complex L1 grammatical patterns in their version of the L2, particularly in cases of natural SLA and creole formation. The concept of imposition, that is, the use of the language-production

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procedures of the dominant language in producing the less dominant one, offers a compelling account of how such replication takes place.

Another concept that can be explained in terms of imposition is that of "contact-induced grammaticalization"—a term introduced by Heine and Kuteva to explain the replication of grammatical functions and categories in cases of language contact. Of particular interest is the notion of "replica grammaticalization," in which "the model language provides a model for both a category and the way that category is replicated" (Heine and Kuteva 2005: 80). An example of this is the reanalysis of the verb *kaba* (< Portuguese *acabar* 'finish') as a marker of Completive aspect in the Surinamese creoles.

(13)	Sranan	A	kownu	doro	kaba
		DET	king	arrive	COMPL
		'The king has arrived.'			

Winford and Migge (2007: 84) demonstrate that the category is closely modeled on the Completive aspect category found in Gbe languages, as illustrated in the following example.

(14)	Ajagbe	àxɔsu	lɔ	à,	e	vá	lɔ	vɔ
		King	DET	TOP	he	come	arrive	COMPL
		'As for the king, he has already arrived.'						

Once more, such kinds of reanalysis, which are abundant in both creoles and other contact languages, can be explained in terms of imposition. In this case, learners established an interlingual identification between the lexical item *kaba* and its Gbe counterpart *vɔ*, on the basis of their semantic similarity. Then they extended the aspectual function of the substrate lexical item to its superstrate counterpart, by simply transferring the lemma of the former to the latter. Other approaches have explained this process in terms of "relexification" and "reanalysis" (Lefebvre 1998), or "functional transfer" (Siegel 2008). Such explanations are quite compatible with the notion of imposition as described here, but the notion of imposition captures the psycholinguistic process involved in these types of grammaticalization more clearly by linking it more directly to the language-production procedure.

Van Coetsem's framework provides a sound basis on which to integrate psycholinguistic approaches to language contact more fully with linguistic approaches, since it is compatible with psycholinguistic models of language production. These links are already well established in Levelt's language production mode, which draws on lexicalist approaches to syntax such as Lexical Functional Grammar (Bresnan 1982). This is in keeping with Levelt's view that "the lexicon is an essential mediator between conceptualization and grammatical encoding" (1989: 51). But in principle the approach is also compatible with other linguistic frameworks. A task for future research is to pursue the connections between linguistic and psycholinguistic approaches to language contact.

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2.7 Conclusion

The unified theory of language contact that we are seeking must integrate the sociolinguistic, the linguistic, and the psycholinguistic aspects of change. First, it must be able to describe and classify both the processes and outcomes of contact-induced change in accurate linguistic terms. Second, it must account for the mechanisms involved in the origins of innovations in individual idiolects—a fundamental aspect of the actuation question that can only be understood in psycholinguistic terms. Third, it must account for the role of macro-level social factors in contact-induced change, as well as the social processes by which change is propagated through social networks, leading to conventionalization of community grammars. It is only by combining these approaches that we will achieve the kind of interdisciplinary model that Weinreich conceived of.

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